Software Requirements Specification (SRS)

HRMS Dashboard System - Comprehensive Specification

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Classification: Industry-Grade Business Application

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive technical specification for the HRMS Dashboard System - an enterprise-grade, AI-powered business management platform designed to unify operations across all organizational departments through intelligent automation, real-time analytics, and seamless workflow integration.

1.2 Scope

The HRMS Dashboard System encompasses six specialized dashboards serving different organizational roles:

- **Executive Dashboard** (CEO & Co-founder)
- **HR Dashboard** (Human Resources)
- **Manager Dashboard** (Project & Team Management)
- **Individual Employee Dashboard**
- **Sales & Client Dashboard**
- **Legal & Finance Dashboard**

1.3 System Goals

- **Unified Business Operations**: Single platform for all business processes
- **AI-Driven Automation**: Intelligent task creation, assignment, and optimization

- **Scalable Architecture**: Support for enterprise-level operations
- **Seamless Integration**: Cross-departmental workflow automation

1.4 Definitions and Acronyms

- **HRMS**: Human Resource Management System
- **AI/ML**: Artificial Intelligence/Machine Learning
- **SLA**: Service Level Agreement
- **API**: Application Programming Interface
- **RBAC**: Role-Based Access Control
- **SSO**: Single Sign-On
- **KPI**: Key Performance Indicator

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2. System Overview

2.1 System Context

The HRMS Dashboard System serves as the central nervous system for enterprise operations, providing:

- **Executive Intelligence**: Strategic oversight and decision support
- **Operational Excellence**: Streamlined workflows and resource optimization
- **Employee Empowerment**: Tools for productivity and career development
- **Client Satisfaction**: Transparent project tracking and communication
- **Compliance Management**: Legal and financial oversight

2.2 System Architecture Overview

```
┌──────────────────────────────────────────────────────────────────┐
│                    PRESENTATION LAYER                            │
├──────────────────────────────────────────────────────────────────┤
│  Executive │    HR    | Manager |Individual| Client |Legal/Fin|
│ Dashboard  |Dashboard|Dashboard|Dashboard |Dashboard|Dashboard|
├──────────────────────────────────────────────────────────────────┤
│                     API GATEWAY LAYER                            │
├──────────────────────────────────────────────────────────────────┤
│          BUSINESS LOGIC & AI/ML SERVICES LAYER                  |
├──────────────────────────────────────────────────────────────────┤
│                    DATA ACCESS LAYER                             │
├──────────────────────────────────────────────────────────────────┤
│              DATABASE & EXTERNAL INTEGRATIONS                    │
└──────────────────────────────────────────────────────────────────┘
```

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3. Functional Requirements

3.1 Executive Dashboard (CEO & Co-founder)

3.1.1 High-Level Overview Widgets

REQ-EXE-001: Real-time Company Metrics

- **Active Employees Counter**: Live count of currently working employees
- **Daily Work Hours Tracker**: Total company work hours (current day/week/month)
- **Productivity Index**: AI-calculated productivity metrics across departments

REQ-EXE-002: AI-Powered Voice Assistant Integration

- **Natural Language Queries**: "How many employees are working today?"
- **Voice Commands**: "Show me this month's productivity report"
- **Intelligent Responses**: Context-aware answers with data visualization
- **Multi-language Support**: Support for primary business languages

REQ-EXE-003: Interactive Chat Functions

- **AI Chat Interface**: 24/7 intelligent business assistant
- **Data Fetching Capabilities**: Real-time query processing across all departments
- **Visual Response Generation**: Automatic chart and graph generation
- **Export Functionality**: PDF/Excel export of chat-generated reports

REQ-EXE-004: Comprehensive Analytics Widgets

- **Employee Activity Heatmap**: Visual representation of workforce engagement
- **Department Performance Comparison**: Cross-departmental efficiency metrics
- **Resource Utilization Metrics**: Equipment, software, and human resource usage
- **Financial Health Indicators**: Cash flow, expenses, profit margins
- **Client Satisfaction Scores**: Aggregated client feedback and project ratings

3.1.2 Data Aggregation Requirements

REQ-EXE-005: Multi-Department Data Integration

- **Manager Data**: Project status, team performance, resource requests
- **HR Data**: Employee metrics, attendance, performance reviews
- **Sales Data**: Pipeline status, client interactions, revenue projections
- **Individual Employee Data**: Task completion, productivity scores, availability
- **Client Data**: Project satisfaction, feedback, payment status
- **Legal/Finance Data**: Compliance status, financial health, contract renewals

REQ-EXE-006: Real-time Data Processing

- **Live Updates**: Maximum 30-second refresh intervals for critical metrics
- **Historical Trending**: 1, 3, 6, 12-month comparative analysis
- **Predictive Analytics**: AI-driven forecasting for key business metrics
- **Anomaly Detection**: Automatic alerts for unusual patterns or metrics

3.2 HR Dashboard

3.2.1 Core HR Functionalities

REQ-HR-001: Payroll Management System

- **Automated Payroll Processing**: Integration with attendance and performance data
- **Multi-currency Support**: Global payroll processing capabilities
- **Tax Calculation Engine**: Automated tax deductions and compliance
- **Payslip Generation**: Digital payslip distribution with employee portal access
- **Bonus and Incentive Management**: Performance-based compensation calculation

REQ-HR-002: Attendance Management

- **Multiple Check-in Methods**: Biometric, RFID, mobile app, web portal
- **Geolocation Tracking**: Location-based attendance for remote/field employees

- **Overtime Calculation**: Automated overtime tracking and approval workflow
- **Leave Integration**: Seamless integration with leave management system

REQ-HR-003: Leave Management System

- **Multiple Leave Types**: Annual, sick, maternity, paternity, special, unpaid leave
- **Automated Approval Workflow**: Multi-level approval with email notifications
- **Leave Balance Tracking**: Real-time leave balance calculation and display
- **Holiday Calendar Integration**: Company-wide holiday management
- **Leave Analytics**: Department-wise leave trends and planning insights

REQ-HR-004: Employee Records Management

- **Comprehensive Employee Profiles**: Personal, professional, and emergency contact information
- **Document Management**: Digital storage of contracts, certificates, and legal documents
- **Performance History**: Complete performance review and rating history
- **Skills Matrix**: Detailed skill tracking and competency mapping
- **Career Progression Tracking**: Promotion history and career development plans

3.2.2 Performance Tracking and Analytics

REQ-HR-005: Individual Performance Monitoring

- **Performance Grading System**: Standardized performance evaluation metrics
- **Goal Setting and Tracking**: Individual and team goal management
- **360-Degree Feedback**: Multi-source feedback collection and analysis
- **Performance Improvement Plans**: Structured improvement planning and monitoring
- **Recognition and Rewards**: Achievement tracking and reward management

REQ-HR-006: Manager Performance Oversight

- **Team Performance Metrics**: Aggregate team performance visualization
- **Manager Effectiveness Scores**: Leadership and management capability assessment
- **Resource Management Efficiency**: Team resource utilization and optimization
- **Project Delivery Success**: Project completion rates and quality metrics
- **Employee Satisfaction Metrics**: Team satisfaction and engagement scores

REQ-HR-007: AI-Powered HR Analytics

- **Employee Engagement Insights**: Predictive analysis of employee satisfaction
- **Attrition Risk Assessment**: AI-driven identification of flight risk employees
- **Recruitment Analytics**: Hiring success rates and candidate quality metrics
- **Training Needs Analysis**: Skill gap identification and training recommendations
- **Workforce Planning**: Predictive workforce planning and capacity optimization

3.3 Manager Dashboard

3.3.1 Project Management Integration

REQ-MGR-001: Sales-to-Project Workflow

- **Automated Project Creation**: AI-powered project generation from sales pipeline
- **Client Onboarding Integration**: Seamless transition from sales to project delivery
- **Project Template Library**: Industry-specific project templates and frameworks
- **Resource Estimation**: AI-driven resource requirement calculation
- **Budget Allocation**: Automated budget creation based on project scope

- **Client Data Analysis**: Comprehensive client requirement analysis
- **Risk Assessment**: Predictive project risk identification and mitigation
- **Timeline Optimization**: AI-optimized project scheduling and milestone planning
- **Resource Conflict Resolution**: Automated resource scheduling and conflict resolution
- **Performance Prediction**: Project success probability and performance forecasting

3.3.2 Task Management System

REQ-MGR-003: Intelligent Task Creation

- **Handbook-Based Task Generation**: Department-specific task libraries (IT, Sales, Design)
- **AI Task Decomposition**: Automated project breakdown into actionable tasks
- **Standard Operating Procedures**: Integration with company SOPs and best practices
- **Task Template Management**: Reusable task templates with time estimates
- **Dependency Mapping**: Automatic task dependency identification and management

REQ-MGR-004: Smart Resource Allocation

- **Skill-Based Assignment**: AI-powered individual selection based on skill matching
- **Workload Optimization**: Intelligent task distribution considering current workload
- **Availability Analysis**: Real-time employee availability and capacity calculation
- **Performance-Based Matching**: Task assignment based on historical performance
- **Learning Opportunity Identification**: Skill development task recommendations

REQ-MGR-005: Advanced Scheduling and Timeline Management

- **Dynamic Timeline Adjustment**: Real-time timeline updates based on progress
- **Critical Path Analysis**: Automated identification of project critical paths
- **Buffer Time Management**: Intelligent buffer allocation for risk mitigation
- **Milestone Tracking**: Automated milestone progress monitoring and reporting
- **Deadline Alert System**: Proactive deadline and bottleneck notifications

3.3.3 Team Management and Monitoring

REQ-MGR-006: Real-time Team Analytics

- **Individual Workload Tracking**: Comprehensive workload visualization per employee
- **Productivity Metrics**: Real-time productivity scoring and trend analysis
- **Skill Utilization Analysis**: Optimal skill deployment and development tracking
- **Team Collaboration Metrics**: Communication and collaboration effectiveness measurement
- **Performance Benchmarking**: Individual and team performance comparison

REQ-MGR-007: Pipeline and Capacity Management

- **Project Pipeline Visualization**: Complete project pipeline overview and forecasting
- **Resource Capacity Planning**: Long-term resource planning and allocation
- **Workload Balancing**: Automated workload distribution across team members
- **Scalability Planning**: Team expansion and contraction planning support
- **Cross-Project Resource Sharing**: Resource optimization across multiple projects

3.4 Individual Employee Dashboard

3.4.1 Personal Productivity Management

REQ-IND-001: Task and Project Management

- **Priority Management**: AI-assisted task prioritization and scheduling
- **Time Tracking**: Integrated time tracking with automatic task association
- **Progress Reporting**: Real-time progress updates and completion tracking
- **Goal Setting**: Personal and professional goal management

REQ-IND-002: Skill Development and Career Management

- **Skill Profile Management**: Comprehensive skill inventory and proficiency tracking
- **Learning Path Recommendations**: AI-suggested training and development opportunities
- **Certification Tracking**: Professional certification and achievement management
- **Career Progression Planning**: Personalized career development roadmaps
- **Mentorship Integration**: Mentor-mentee matching and progress tracking

REQ-IND-003: Work-Life Balance Tools

- **Schedule Management**: Personal calendar integration with work assignments
- **Availability Setting**: Flexible availability and working hours management
- **Workload Monitoring**: Personal workload alerts and optimization suggestions
- **Wellness Tracking**: Basic wellness and stress level monitoring
- **Break and Rest Reminders**: Automated wellness and productivity reminders

3.4.2 Communication and Collaboration

REQ-IND-004: Team Communication Interface

- **Project Communication**: Integrated project-specific communication channels
- **Status Updates**: Automated and manual status update capabilities
- **Feedback System**: Peer and manager feedback collection and display
- **Knowledge Sharing**: Access to company knowledge base and documentation
- **Collaboration Tools**: File sharing, document collaboration, and version control

3.5 Sales & Client Dashboard

3.5.1 Sales Pipeline Management

REQ-SAL-001: Lead and Opportunity Management

- **Lead Tracking System**: Comprehensive lead lifecycle management
- **Opportunity Scoring**: AI-powered opportunity qualification and scoring
- **Sales Funnel Analytics**: Visual sales pipeline with conversion tracking
- **Customer Relationship Management**: Integrated CRM with interaction history
- **Proposal and Quote Management**: Automated proposal generation and tracking

REQ-SAL-002: Client Onboarding Integration

- **Seamless Handoff Process**: Automated transition from sales to project delivery
- **Client Information Transfer**: Complete client data migration to project systems
- **Expectation Setting**: Clear project scope and timeline communication
- **Kickoff Meeting Scheduling**: Automated project kickoff coordination
- **Success Metrics Definition**: Clear success criteria and measurement setup

3.5.2 Client Portal and Communication

REQ-CLT-001: Project Visibility Dashboard

- **Project Progress Tracking**: Real-time project status and milestone completion

- **Task Distribution Visualization**: Detailed task breakdown and assignment view
- **Timeline and Deadline Tracking**: Interactive project timeline with key dates
- **Deliverable Status**: Clear deliverable status and approval workflows

REQ-CLT-002: Communication and Feedback System

- **Direct Communication Channel**: Secure client-team communication platform
- **Feedback Collection**: Structured feedback collection and rating system
- **Change Request Management**: Formal change request submission and tracking
- **Document Sharing**: Secure document exchange and version control
- **Meeting Scheduling**: Integrated meeting scheduling and calendar management

3.6 Legal & Finance Dashboard

3.6.1 Legal Management System

REQ-LEG-001: Contract and Agreement Management

- **Contract Lifecycle Management**: Complete contract creation, approval, and renewal process
- **Compliance Tracking**: Regulatory compliance monitoring and reporting
- **Risk Assessment**: Legal risk identification and mitigation planning
- **Document Management**: Secure legal document storage and retrieval
- **Audit Trail**: Complete legal action and decision audit trail

REQ-LEG-002: Automated Legal Operations

- **Renewal Alert System**: Automated contract and agreement renewal notifications
- **Compliance Monitoring**: Real-time compliance status tracking and reporting
- **Legal Calendar Management**: Important legal date and deadline tracking
- **Template Management**: Legal document template library and customization
- **Approval Workflow**: Multi-level legal document approval processes

3.6.2 Financial Management System

REQ-FIN-001: Comprehensive Financial Operations

- **Accounts Receivable/Payable**: Complete AR/AP management and tracking
- **Expense Management**: Employee expense submission, approval, and reimbursement
- **Budget Management**: Department-wise budget creation, tracking, and variance analysis
- **Invoice Management**: Automated invoice generation, delivery, and payment tracking
- **Financial Reporting**: Comprehensive financial reporting and analytics

REQ-FIN-002: Advanced Financial Analytics

- **Cash Flow Forecasting**: AI-powered cash flow prediction and planning
- **Profitability Analysis**: Project and department profitability assessment
- **Cost Center Management**: Detailed cost allocation and center performance
- **Revenue Recognition**: Automated revenue recognition and reporting
- **Financial KPI Dashboard**: Real-time financial health indicators and metrics

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4. Non-Functional Requirements

4.1 Performance Requirements

- **Widget Refresh**: Maximum 2 seconds for widget data refresh
- **Search Operations**: Maximum 1 second for search result display
- **Report Generation**: Maximum 30 seconds for complex report generation
- **AI Processing**: Maximum 10 seconds for AI analysis and recommendations

REQ-PERF-002: Throughput Requirements

- **Concurrent Users**: Support for minimum 1,000 concurrent users
- **API Calls**: Handle minimum 10,000 API calls per minute
- **Data Processing**: Process minimum 1 million records per hour
- **File Upload**: Support files up to 100MB with concurrent uploads
- **Database Operations**: Handle minimum 1,000 database transactions per second

4.2 Scalability Requirements

REQ-SCAL-001: Horizontal Scaling

- **Auto-scaling**: Automatic server scaling based on load
- **Load Balancing**: Intelligent load distribution across server instances
- **Database Sharding**: Support for database horizontal partitioning
- **CDN Integration**: Content delivery network for global performance
- **Microservices Architecture**: Independently scalable service components

REQ-SCAL-002: Data Scaling

- **Storage Capacity**: Support for minimum 10TB of structured data
- **User Scaling**: Support for up to 50,000 registered users
- **Transaction Volume**: Handle up to 10 million transactions per day
- **Historical Data**: Maintain 7 years of historical data accessibility
- **Backup and Archive**: Automated data backup and archival processes

4.3 Availability Requirements

REQ-AVAIL-001: System Uptime

- **Service Level Agreement**: 99.9% system availability (< 8.77 hours downtime/year)
- **Planned Maintenance**: Maximum 4 hours monthly maintenance window
- **Disaster Recovery**: Maximum 4-hour recovery time objective (RTO)
- **Data Recovery**: Maximum 1-hour recovery point objective (RPO)
- **Health Monitoring**: 24/7 system health monitoring and alerting

4.4 Security Requirements

REQ-SEC-001: Authentication and Authorization

- **Multi-Factor Authentication**: Mandatory MFA for all user accounts
- **Single Sign-On**: Integration with enterprise SSO solutions
- **Role-Based Access Control**: Granular permission management
- **Session Management**: Secure session handling with automatic timeout
- **Password Policy**: Enforced strong password requirements

REQ-SEC-002: Data Security

- **Encryption at Rest**: AES-256 encryption for stored data
- **Encryption in Transit**: TLS 1.3 for all data transmission
- **API Security**: OAuth 2.0 and JWT token-based API authentication
- **Data Masking**: Sensitive data masking in non-production environments

5. System Architecture

5.1 Architecture Overview

The HRMS Dashboard System follows a microservices architecture pattern with the following key components:

```
┌─────────────────────────────────────────────────────────────┐
│                      CLIENT LAYER                           │
│  Web App │ Mobile App │ Voice Interface │ API Clients       │
├─────────────────────────────────────────────────────────────┤
│                   API GATEWAY & LOAD BALANCER               │
│        Authentication │ Rate Limiting │ Request Routing     │
├─────────────────────────────────────────────────────────────┤
│                     MICROSERVICES LAYER                     │
│ User Mgmt │ HR │ Project │ Finance │ AI/ML │ Notification   │
├─────────────────────────────────────────────────────────────┤
│                      MESSAGING & EVENTS                     │
│          Message Queue │ Event Bus │ Real-time Updates      │
├─────────────────────────────────────────────────────────────┤
│                        DATA LAYER                           │
│   Primary DB │ Cache │ Search │ File Storage │ AI Models     │
├─────────────────────────────────────────────────────────────┤
│                   INFRASTRUCTURE LAYER                      │
│    Container Orchestration │ Monitoring │ Logging │ CI/CD   │
└─────────────────────────────────────────────────────────────┘
```

5.2 Technology Stack

5.2.1 Frontend Technologies

- **Web Framework**: React.js 18+ with TypeScript
- **State Management**: Redux Toolkit with RTK Query
- **UI Component Library**: Material-UI (MUI) with custom theming
- **Charts and Visualization**: D3.js, Chart.js, and custom dashboards
- **Voice Integration**: Web Speech API with custom voice processing
- **Mobile Development**: React Native for mobile applications

5.2.2 Backend Technologies

- **API Framework**: Node.js with Express.js and TypeScript
- **Authentication**: Auth0 or custom JWT-based authentication
- **Database**: PostgreSQL (primary), Redis (caching), Elasticsearch (search)
- **Message Queue**: Apache Kafka for event streaming
- **File Storage**: Amazon S3 or Azure Blob Storage
- **Container Platform**: Docker with Kubernetes orchestration

5.2.3 AI/ML Technologies

- **Machine Learning**: Python with TensorFlow/PyTorch
- **Natural Language Processing**: OpenAI GPT integration
- **Voice Processing**: Google Cloud Speech-to-Text and Text-to-Speech
- **Recommendation Engine**: Custom collaborative filtering algorithms
- **Predictive Analytics**: Apache Spark with MLlib

- **Cloud Platform**: AWS, Azure, or Google Cloud Platform
- **Container Orchestration**: Kubernetes with Helm charts
- **CI/CD Pipeline**: Jenkins, GitLab CI, or GitHub Actions
- **Monitoring**: Prometheus, Grafana, and ELK Stack
- **Security**: HashiCorp Vault for secrets management

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6. Database Design

6.1 Database Architecture

The system uses a polyglot persistence approach with different databases optimized for specific use cases:

6.1.1 Primary Database (PostgreSQL)

Core Business Data

```sql
-- Users and Authentication
CREATE TABLE users (
    user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    username VARCHAR(50) UNIQUE NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    role_id UUID REFERENCES roles(role_id),
    department_id UUID REFERENCES departments(department_id),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    is_active BOOLEAN DEFAULT true
);

-- Employee Information
CREATE TABLE employees (
    employee_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(user_id),
    employee_code VARCHAR(20) UNIQUE NOT NULL,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    date_of_birth DATE,
    hire_date DATE NOT NULL,
    job_title VARCHAR(100),
    salary DECIMAL(12,2),
    manager_id UUID REFERENCES employees(employee_id),
    skills JSONB,
    performance_rating DECIMAL(3,2),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Projects and Tasks
CREATE TABLE projects (
    project_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_name VARCHAR(255) NOT NULL,
    client_id UUID REFERENCES clients(client_id),
    manager_id UUID REFERENCES employees(employee_id),
    start_date DATE,
    end_date DATE,
    status VARCHAR(50) DEFAULT 'ACTIVE',
    budget DECIMAL(15,2),
    description TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
```

```sql
CREATE TABLE tasks (
    task_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_id UUID REFERENCES projects(project_id),
    assigned_to UUID REFERENCES employees(employee_id),
    task_name VARCHAR(255) NOT NULL,
    description TEXT,
    estimated_hours INTEGER,
    actual_hours INTEGER,
    priority VARCHAR(20) DEFAULT 'MEDIUM',
    status VARCHAR(50) DEFAULT 'TODO',
    due_date DATE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

6.1.2 Time Series Database (InfluxDB)

Performance Metrics and Analytics

```sql
-- Employee Performance Metrics
CREATE MEASUREMENT employee_performance (
    time TIMESTAMP,
    employee_id STRING,
    productivity_score FLOAT,
    tasks_completed INTEGER,
    hours_worked FLOAT,
    utilization_rate FLOAT
);

-- System Usage Metrics
CREATE MEASUREMENT system_usage (
    time TIMESTAMP,
    user_id STRING,
    dashboard_type STRING,
    session_duration INTEGER,
    actions_performed INTEGER
);
```

6.2 Data Models

6.2.1 User Management Model

```typescript
interface User {
  userId: string;
  username: string;
  email: string;
  role: UserRole;
  department: Department;
  profile: UserProfile;
  permissions: Permission[];
  preferences: UserPreferences;
  createdAt: Date;
  updatedAt: Date;
  isActive: boolean;
}

interface UserRole {
  roleId: string;
  roleName: string;
  permissions: Permission[];
```

```
}
```

6.2.2 Project Management Model

```
interface Project {
  projectId: string;
  projectName: string;
  client: Client;
  manager: Employee;
  team: Employee[];
  tasks: Task[];
  timeline: ProjectTimeline;
  budget: ProjectBudget;
  status: ProjectStatus;
  progress: ProjectProgress;
  aiInsights: AIInsights;
}

interface Task {
  taskId: string;
  projectId: string;
  assignedTo: Employee;
  taskName: string;
  description: string;
  estimatedHours: number;
  actualHours: number;
  priority: TaskPriority;
  status: TaskStatus;
  dependencies: Task[];
  aiRecommendations: AIRecommendations;
}
```

---

7. API Specifications

7.1 RESTful API Design

The system follows RESTful API design principles with OpenAPI 3.0 specification.

7.1.1 Authentication APIs

```
// User Authentication
POST /api/v1/auth/login
Request: {
  username: string;
  password: string;
  mfaCode?: string;
}
Response: {
  accessToken: string;
  refreshToken: string;
  user: UserProfile;
  permissions: Permission[];
}

// Token Refresh
POST /api/v1/auth/refresh
Request: {
  refreshToken: string;
}
Response: {
```

```
  refreshToken: string;
}
```

7.1.2 Executive Dashboard APIs

```
// Company Metrics
GET /api/v1/executive/metrics
Query Parameters: {
  timeframe?: 'day' | 'week' | 'month' | 'quarter';
  departments?: string[];
}
Response: {
  activeEmployees: number;
  totalWorkHours: number;
  productivityScore: number;
  revenueMetrics: RevenueData;
  departmentMetrics: DepartmentMetric[];
}

// AI Voice Query
POST /api/v1/executive/voice-query
Request: {
  audioData: string; // base64 encoded
  language?: string;
}
Response: {
  transcription: string;
  response: string;
  visualData?: ChartData;
  audioResponse: string; // base64 encoded
}
```

7.1.3 Manager Dashboard APIs

```
// Project Management
GET /api/v1/manager/projects
Query Parameters: {
  status?: ProjectStatus;
  managerId?: string;
  limit?: number;
  offset?: number;
}
Response: {
  projects: Project[];
  totalCount: number;
  aiInsights: ProjectInsights;
}

// Task Assignment
POST /api/v1/manager/tasks/assign
Request: {
  projectId: string;
  taskDetails: TaskDetails;
  aiOptimization: boolean;
}
Response: {
  task: Task;
  recommendedAssignee: Employee;
  optimizationScore: number;
  alternativeAssignees: Employee[];
}
```

7.2.1 Real-time Dashboard Updates

```
// WebSocket Connection
ws://api.domain.com/ws/dashboard/{dashboardType}

// Message Types
interface DashboardUpdate {
  type: 'METRIC_UPDATE' | 'TASK_STATUS' | 'NOTIFICATION';
  data: any;
  timestamp: Date;
}

// Subscription Topics
{
  'executive.metrics': ExecutiveMetrics;
  'hr.attendance': AttendanceUpdate;
  'manager.tasks': TaskUpdate;
  'individual.notifications': Notification;
}
```

7.3 AI/ML APIs

7.3.1 Task Optimization API

```
POST /api/v1/ai/optimize-task-assignment
Request: {
  projectRequirements: ProjectRequirements;
  availableEmployees: Employee[];
  constraints: AssignmentConstraints;
}
Response: {
  optimizedAssignment: TaskAssignment[];
  confidenceScore: number;
  alternativeOptions: AlternativeAssignment[];
  reasoning: string;
}
```

7.3.2 Predictive Analytics API

```
POST /api/v1/ai/predict-project-outcome
Request: {
  projectId: string;
  currentProgress: ProjectProgress;
  historicalData: boolean;
}
Response: {
  completionProbability: number;
  estimatedCompletionDate: Date;
  riskFactors: RiskFactor[];
  recommendations: Recommendation[];
}
```

---

8. Security Requirements

8.1 Authentication and Authorization

REQ-SEC-AUTH-001: MFA Implementation

- **Primary Factor**: Username/password or SSO
- **Secondary Factors**: SMS OTP, email OTP, authenticator app, biometric
- **Backup Methods**: Recovery codes, admin override
- **Risk-Based Authentication**: IP-based and behavior-based risk assessment
- **Session Security**: Secure session tokens with automatic expiration

8.1.2 Role-Based Access Control (RBAC)

REQ-SEC-AUTH-002: Granular Permissions

```
interface Permission {
  resource: string; // 'projects', 'employees', 'reports'
  action: string;   // 'read', 'write', 'delete', 'approve'
  scope: string;    // 'own', 'team', 'department', 'company'
  conditions?: PermissionCondition[];
}

interface Role {
  roleId: string;
  roleName: string;
  permissions: Permission[];
  hierarchyLevel: number;
  inheritedRoles: Role[];
}
```

8.2 Data Protection

8.2.1 Encryption Standards

REQ-SEC-DATA-001: Encryption Requirements

- **Data at Rest**: AES-256 encryption for all stored data
- **Data in Transit**: TLS 1.3 for all client-server communication
- **Key Management**: Hardware Security Module (HSM) or cloud KMS
- **Certificate Management**: Automated certificate rotation and renewal
- **Database Encryption**: Transparent Data Encryption (TDE) for databases

8.2.2 Data Privacy and Compliance

REQ-SEC-DATA-002: Privacy Protection

- **Data Minimization**: Collect only necessary personal information
- **Purpose Limitation**: Use data only for specified business purposes
- **Retention Policies**: Automated data purging based on retention schedules
- **Right to Erasure**: Complete data deletion on user request
- **Data Portability**: Export user data in standard formats

8.3 Application Security

8.3.1 Input Validation and Sanitization

REQ-SEC-APP-001: Input Security

- **XSS Protection**: Input sanitization and Content Security Policy
- **CSRF Protection**: Anti-CSRF tokens for state-changing operations
- **File Upload Security**: File type validation and malware scanning
- **API Rate Limiting**: Request throttling and DDoS protection

8.3.2 Security Monitoring and Incident Response

REQ-SEC-MON-001: Security Operations

- **SIEM Integration**: Security Information and Event Management
- **Threat Detection**: Anomaly detection and behavioral analysis
- **Incident Response**: Automated incident response workflows
- **Forensic Logging**: Immutable audit trails for security events
- **Vulnerability Management**: Regular security scanning and patching

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9. Integration Requirements

9.1 Third-Party Integrations

9.1.1 Authentication Providers

REQ-INT-AUTH-001: SSO Integration

- **Active Directory**: LDAP/SAML integration for enterprise authentication
- **OAuth 2.0 Providers**: Google, Microsoft, LinkedIn integration
- **SAML 2.0**: Support for enterprise SAML identity providers
- **OpenID Connect**: Standard OIDC implementation
- **Custom Identity Providers**: Flexible integration framework

9.1.2 Communication Platforms

REQ-INT-COMM-001: Communication Integration

- **Email Services**: SMTP integration with Gmail, Outlook, SendGrid
- **Slack Integration**: Real-time notifications and bot interactions
- **Microsoft Teams**: Meeting scheduling and notification integration
- **WhatsApp Business**: Client communication and status updates
- **Video Conferencing**: Zoom, Google Meet, Microsoft Teams integration

9.1.3 Financial Systems

REQ-INT-FIN-001: Accounting Integration

- **QuickBooks**: Automated accounting data synchronization
- **SAP Integration**: Enterprise resource planning integration
- **Banking APIs**: Payment processing and transaction monitoring
- **Invoice Systems**: Automated invoice generation and tracking
- **Expense Management**: Receipt processing and expense categorization

9.2 API Integration Framework

9.2.1 Integration Architecture

```
interface IntegrationConfig {
  providerId: string;
  providerName: string;
  authConfig: AuthenticationConfig;
  endpoints: APIEndpoint[];
  dataMapping: DataMapping[];
  syncSchedule: SyncSchedule;
  errorHandling: ErrorHandlingConfig;
}

interface DataMapping {
  sourceField: string;
  targetField: string;
  transformation?: TransformationRule[];
  validation?: ValidationRule[];
}
```

9.2.2 Webhook Management

REQ-INT-WEB-001: Webhook Framework

- **Webhook Registration**: Dynamic webhook endpoint registration
- **Event Filtering**: Selective event subscription and filtering
- **Retry Logic**: Exponential backoff for failed webhook deliveries
- **Security**: HMAC signature verification for webhook authenticity
- **Monitoring**: Webhook delivery monitoring and alerting

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10. User Interface Requirements

10.1 Design System and Standards

10.1.1 Design Principles

REQ-UI-DES-001: User Experience Standards

- **Consistency**: Unified design language across all dashboards
- **Accessibility**: WCAG 2.1 AA compliance for inclusive design
- **Responsiveness**: Mobile-first responsive design approach
- **Performance**: Maximum 3-second page load times
- **Internationalization**: Multi-language support with RTL compatibility

10.1.2 Visual Design System

REQ-UI-VIS-001: Design Components

```
interface DesignSystem {
  colors: {
    primary: ColorPalette;
    secondary: ColorPalette;
    semantic: SemanticColors; // success, warning, error, info
    neutral: NeutralColors;
  };
  typography: {
    fontFamilies: FontFamily[];
    scales: TypographyScale;
    weights: FontWeight[];
  };
  spacing: SpacingScale;
```

```
  layouts: LayoutTemplates;
}
```

10.2 Dashboard-Specific UI Requirements

10.2.1 Executive Dashboard UI

REQ-UI-EXE-001: Executive Interface Design

- **Widget-Based Layout**: Customizable dashboard widgets with drag-and-drop
- **Voice Interface**: Visual voice recognition with speech-to-text display
- **Data Visualization**: Interactive charts with drill-down capabilities
- **Mobile Optimization**: Executive mobile app with key metrics
- **Dark Mode**: Professional dark theme for extended usage

10.2.2 Manager Dashboard UI

REQ-UI-MGR-001: Project Management Interface

- **Kanban Boards**: Interactive task management with drag-and-drop
- **Gantt Charts**: Project timeline visualization with dependencies
- **Resource Views**: Team workload visualization and capacity planning
- **AI Recommendations**: Prominent display of AI-generated insights
- **Collaboration Tools**: Integrated communication and file sharing

10.2.3 Mobile Application Requirements

REQ-UI-MOB-001: Mobile Interface Standards

- **Native Performance**: Native mobile app performance on iOS and Android
- **Offline Capability**: Critical functionality available offline
- **Push Notifications**: Real-time notifications with customizable preferences
- **Biometric Authentication**: Fingerprint and face recognition support
- **Quick Actions**: One-tap access to frequently used features

10.3 Accessibility Requirements

10.3.1 Web Accessibility Standards

REQ-UI-ACC-001: Accessibility Compliance

- **WCAG 2.1 AA**: Full compliance with web accessibility guidelines
- **Screen Reader Support**: Comprehensive screen reader compatibility
- **Keyboard Navigation**: Full keyboard navigation support
- **Color Contrast**: Minimum 4.5:1 contrast ratio for text
- **Focus Management**: Clear focus indicators and logical tab order

10.3.2 Assistive Technology Support

REQ-UI-ASS-001: Assistive Technology Integration

- **Voice Control**: Integration with voice control systems
- **High Contrast Mode**: Enhanced contrast options for visual impairments
- **Text Scaling**: Support for 200% text scaling without horizontal scrolling
- **Alternative Text**: Comprehensive alt text for all visual elements

11. AI/ML Requirements

11.1 AI/ML Architecture

11.1.1 Machine Learning Pipeline

REQ-AI-PIPE-001: ML Pipeline Architecture

```
Data Collection → Data Preprocessing → Feature Engineering →
Model Training → Model Validation → Model Deployment →
Performance Monitoring → Model Retraining
```

11.1.2 AI Service Components

REQ-AI-COMP-001: AI Service Architecture

```typescript
interface AIService {
  serviceName: string;
  modelType: 'classification' | 'regression' | 'nlp' | 'recommendation';
  inputSchema: DataSchema;
  outputSchema: DataSchema;
  performanceMetrics: ModelMetrics;
  trainingData: TrainingDataset;
  version: string;
  deploymentStatus: DeploymentStatus;
}
```

11.2 Specific AI/ML Capabilities

11.2.1 Natural Language Processing

REQ-AI-NLP-001: Language Understanding

- **Voice Recognition**: Multi-language speech-to-text with 95% accuracy
- **Intent Classification**: Business query intent understanding
- **Entity Extraction**: Automatic extraction of business entities from text
- **Sentiment Analysis**: Employee and client sentiment analysis
- **Text Generation**: Automated report and summary generation

11.2.2 Predictive Analytics

REQ-AI-PRED-001: Business Predictions

- **Project Success Prediction**: 85% accuracy in project outcome prediction
- **Employee Attrition Risk**: Early identification of flight risk employees
- **Resource Demand Forecasting**: Accurate resource requirement prediction
- **Performance Prediction**: Individual and team performance forecasting
- **Financial Forecasting**: Revenue and expense prediction models

11.2.3 Optimization Algorithms

REQ-AI-OPT-001: Intelligent Optimization

- **Resource Allocation**: Efficient resource distribution algorithms
- **Schedule Optimization**: Intelligent meeting and deadline scheduling
- **Workload Balancing**: Dynamic workload distribution optimization
- **Cost Optimization**: Budget allocation and expense optimization

11.3 AI/ML Data Requirements

11.3.1 Training Data Management

REQ-AI-DATA-001: Data Pipeline

- **Data Quality**: Automated data quality assessment and cleaning
- **Feature Store**: Centralized feature repository for ML models
- **Data Versioning**: Version control for training datasets
- **Synthetic Data**: Generation of synthetic data for model training
- **Privacy Preservation**: Differential privacy for sensitive data

11.3.2 Model Management

REQ-AI-MODEL-001: MLOps Pipeline

- **Model Registry**: Centralized model repository with version control
- **A/B Testing**: Automated model performance comparison
- **Continuous Learning**: Online learning and model adaptation
- **Model Monitoring**: Real-time model performance monitoring
- **Automated Retraining**: Scheduled model retraining based on performance drift

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12. Performance Requirements

12.1 System Performance Specifications

12.1.1 Response Time Requirements

REQ-PERF-RESP-001: Response Time Standards | Operation Type | Target Response Time | Maximum Acceptable | |----------|----------|----------| | Dashboard Load | < 2 seconds | < 5 seconds | | Widget Refresh | < 1 second | < 3 seconds | | Search Query | < 0.5 seconds | < 2 seconds | | Report Generation | < 10 seconds | < 30 seconds | | AI Analysis | < 5 seconds | < 15 seconds | | File Upload | < 3 seconds | < 10 seconds |

12.1.2 Throughput Requirements

REQ-PERF-THRU-001: System Throughput Standards

- **Concurrent Users**: Support 5,000 concurrent active users
- **API Throughput**: Handle 50,000 requests per minute
- **Database Operations**: Process 5,000 transactions per second
- **Real-time Updates**: Deliver updates to 10,000 concurrent connections
- **File Processing**: Process 1,000 file uploads simultaneously

12.2 Scalability Requirements

12.2.1 Horizontal Scaling

REQ-PERF-SCALE-001: Auto-scaling Capabilities

- **Database Scaling**: Read replicas and connection pooling
- **Cache Layer**: Distributed caching with Redis Cluster
- **Load Balancing**: Intelligent load distribution across instances
- **CDN Integration**: Global content delivery for static assets

12.2.2 Performance Optimization

REQ-PERF-OPT-001: Optimization Strategies

- **Database Optimization**: Query optimization and index management
- **Caching Strategy**: Multi-level caching (browser, CDN, application, database)
- **Code Optimization**: Performance profiling and optimization
- **Asset Optimization**: Image compression and minification
- **Lazy Loading**: Progressive loading of dashboard components

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13. Deployment Requirements

13.1 Deployment Architecture

13.1.1 Container Orchestration

REQ-DEP-CONT-001: Kubernetes Deployment

Kubernetes Deployment Configuration apiVersion: apps/v1 kind: Deployment metadata: name: hrms-dashboard labels: app: hrms-dashboard spec: replicas: 3 selector: matchLabels: app: hrms-dashboard template: metadata: labels: app: hrms-dashboard spec: containers: - name: hrms-api image: hrms/api:latest ports: - containerPort: 3000 env: - name: DATABASE_URL valueFrom: secretKeyRef: name: hrms-secrets key: database-url resources: requests: memory: "512Mi" cpu: "250m" limits: memory: "1Gi" cpu: "500m"

13.1.2 Multi-Environment Deployment

REQ-DEP-ENV-001: Environment Management

- **Development**: Local development with Docker Compose
- **Staging**: Production-like environment for testing
- **Production**: High-availability production deployment
- **Disaster Recovery**: Geographically distributed backup deployment
- **Blue-Green Deployment**: Zero-downtime deployment strategy

13.2 Infrastructure Requirements

13.2.1 Cloud Infrastructure

REQ-DEP-INFRA-001: Cloud Architecture

- **Compute**: Auto-scaling virtual machines or containers
- **Storage**: High-performance SSD storage with automatic backups
- **Networking**: Virtual private cloud with security groups
- **Load Balancing**: Application and network load balancers
- **DNS**: Global DNS with health checks and failover

13.2.2 Database Deployment

- **Primary Database**: PostgreSQL cluster with read replicas
- **Cache Layer**: Redis cluster for session and application caching
- **Search Engine**: Elasticsearch cluster for full-text search
- **Backup Strategy**: Automated daily backups with point-in-time recovery
- **Monitoring**: Database performance monitoring and alerting

13.3 DevOps and CI/CD

13.3.1 Continuous Integration

REQ-DEP-CI-001: CI Pipeline

CI/CD Pipeline Configuration stages: - code-quality - unit-tests - integration-tests - security-scan - build - deploy-staging - e2e-tests - deploy-production code-quality: script: - npm run lint - npm run type-check - sonar-scanner unit-tests: script: - npm run test:unit - npm run coverage coverage: '/Lines\s*:\s*(\d+\.\d+)%/' security-scan: script: - npm audit - docker run --rm -v "$PWD:/app" securecodewarrior/dependency-check

13.3.2 Deployment Automation

REQ-DEP-AUTO-001: Automated Deployment

- **Infrastructure as Code**: Terraform or CloudFormation templates
- **Configuration Management**: Ansible or Helm for configuration
- **Secret Management**: HashiCorp Vault or cloud secret managers
- **Database Migrations**: Automated schema migrations
- **Health Checks**: Comprehensive health check endpoints

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14. Testing Requirements

14.1 Testing Strategy

14.1.1 Test Pyramid Structure

REQ-TEST-STRAT-001: Testing Levels

```
                    E2E Tests (10%)
                   /               \
             Integration Tests (20%)
            /                       \
        Unit Tests (70%)
```

14.1.2 Testing Framework

REQ-TEST-FRAME-001: Testing Tools and Frameworks

- **Unit Testing**: Jest for JavaScript/TypeScript, pytest for Python
- **Integration Testing**: Supertest for API testing, Testcontainers for database testing
- **End-to-End Testing**: Playwright or Cypress for UI automation
- **Performance Testing**: K6 or Artillery for load testing
- **Security Testing**: OWASP ZAP for security scanning

14.2 Specific Testing Requirements

REQ-TEST-FUNC-001: Functional Test Coverage

- **API Testing**: 100% endpoint coverage with positive and negative test cases
- **UI Testing**: Critical user journey automation (login, dashboard navigation, core features)
- **Integration Testing**: Third-party service integration validation
- **Data Validation**: Input validation and data integrity testing
- **Business Logic**: Core business rule validation and edge case testing

14.2.2 Non-Functional Testing

REQ-TEST-NONFUNC-001: Performance and Security Testing

- **Load Testing**: 5,000 concurrent users simulation
- **Stress Testing**: System behavior under extreme load conditions
- **Security Testing**: Penetration testing and vulnerability assessment
- **Accessibility Testing**: WCAG 2.1 compliance validation
- **Browser Compatibility**: Testing across major browsers and devices

14.2.3 AI/ML Testing

REQ-TEST-AI-001: AI Model Testing

- **Model Accuracy**: Validation of AI prediction accuracy and performance
- **Bias Testing**: Detection and mitigation of algorithmic bias
- **Data Quality**: Training data quality and integrity validation
- **Model Drift**: Continuous monitoring for model performance degradation
- **Adversarial Testing**: Robustness testing against adversarial inputs

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15. Appendices

15.1 Glossary of Terms

Term	Definition
Dashboard	A visual interface displaying key business metrics and controls
Widget	A modular UI component displaying specific data or functionality
Microservice	An independently deployable service component
API Gateway	A centralized entry point for all client requests
RBAC	Role-Based Access Control for managing user permissions
SSO	Single Sign-On for unified authentication across systems
CI/CD	Continuous Integration and Continuous Deployment
MLOps	Machine Learning Operations for model lifecycle management

15.2 Compliance and Standards

- **GDPR**: General Data Protection Regulation compliance
- **CCPA**: California Consumer Privacy Act compliance
- **SOX**: Sarbanes-Oxley Act compliance for financial reporting
- **HIPAA**: Health Insurance Portability and Accountability Act (if applicable)
- **ISO 27001**: Information Security Management System standards

15.2.2 Technical Standards

- **OpenAPI 3.0**: API documentation and specification standard
- **OAuth 2.0**: Authorization framework standard
- **JWT**: JSON Web Token standard for secure token transmission
- **WCAG 2.1**: Web Content Accessibility Guidelines
- **W3C Standards**: HTML5, CSS3, and JavaScript standards compliance

15.3 Risk Assessment Matrix

Risk Category	Probability	Impact	Mitigation Strategy
Data Breach	Medium	High	Multi-layer security, encryption, monitoring
System Downtime	Low	High	Redundancy, disaster recovery, monitoring
Performance Degradation	Medium	Medium	Auto-scaling, performance monitoring
AI Model Bias	Medium	Medium	Bias testing, diverse training data
Integration Failure	High	Medium	Comprehensive testing, fallback mechanisms

15.4 Success Metrics and KPIs

15.4.1 Technical KPIs

- **System Uptime**: 99.9% availability target
- **Response Time**: <3 seconds for 95% of requests
- **User Satisfaction**: >4.5/5 user satisfaction score
- **Bug Rate**: <1 critical bug per release
- **Security Incidents**: Zero data breaches

15.4.2 Business KPIs

- **User Adoption**: 90% user adoption within 6 months
- **Productivity Improvement**: 25% increase in operational efficiency
- **Cost Reduction**: 20% reduction in administrative overhead
- **ROI**: 300% return on investment within 2 years
- **Customer Satisfaction**: >4.7/5 client satisfaction score

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This document represents a comprehensive technical specification for an enterprise-grade HRMS Dashboard System. All requirements should be validated with stakeholders before implementation begins.